Training data

200K examples, 35 classes

FinHeadlineMix headlines

Teacher: Qwen3-32B

label | reasoning trace

Dataset labels and traces are generated from **Qwen3-32B**.

1 Response-only LoRA

LoRA Fine-tuning

Qwen3-8B + LoRA, r = 32

prompt | label

loss

- Create a strong baseline by tuning the hyperparameters.
- Ablate on "*hyperparameters*": **α**, **LR**, **Min LR**, **β_1**, **Warm-up**
- Select top runs → Stage 2

2 Single-adapter Reasoning

Trace + Label, one adapter

Qwen3-8B + LoRA, r = 32

prompt | reasoning trace | label

$1-\lambda$ | λ

loss

$$L = \lambda\, L_{\text{label}} + (1-\lambda)\, L_{\text{trace}}$$

- Ablate on "*different λ values*": **λ ∈ {0.1, 0.3, 0.5, 0.7, 0.9, 1.0}**
- Select top runs → Stage 3

3 Dual-adapter Composite

Trace adapter + Label adapter

Qwen3-8B + 2 LoRAs, r = 32

Trace adapter, r_{trace} — both losses

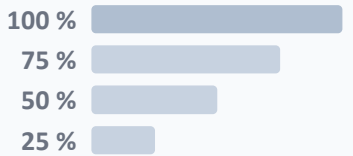
Label adapter, r_{label} — label loss

- Ablate on "*rank splits*": **(16,16), (24,8), (8,24)**
- λ: selected ± adjacent values

4 Summarized Traces

Repeat Stages 2 and 3

Same runs, shorter traces



- Ablate on "*trace length*": **25%, 50%, 75%**
- Teacher summarizes each trace to a target length.

Reproducibility:
Every experiment is repeated with **3 random seeds** → mean ± std is reported.

Evaluation Tasks:
- **In-domain Tasks**: Acc., Macro-F1, Acc. argmax, NLL, Brier
- **Retention Tasks**: MMLU-Pro, GPQA-Diamond, IFEval

Selection criterion:
Keep runs that beat the frozen model on *all* tasks.